



TEMPERATURE



HUMIDITY



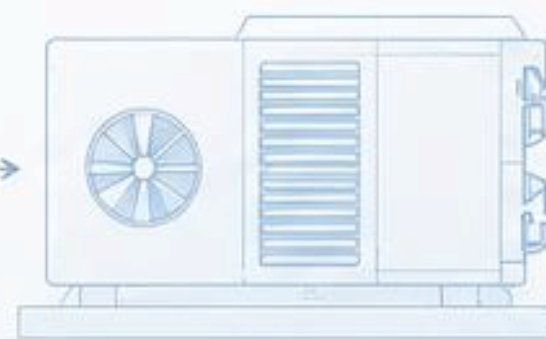
IAQ



ENERGY
EFFICIENCY

Decentralized HVAC Controller Design and Simulation Verification

Team Ventus | Department of Mechanical Engineering
University of Peradeniya | ME325 — Third Year Group Project
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Project Objectives

Objective 1

Control Design: Design per-zone controllers that drive zone temperature, humidity, and CO₂ into user-defined comfort sets while minimizing energy use

Objective 2

Coordination: Implement a lightweight central AHU policy that selects supply conditions and total flow to meet aggregate thermal, moisture, and IAQ loads

Objective 3

Simulation Verification: Demonstrate closed-loop performance and energy savings using a Python based simulation interface that utilizes EnergyPlus as the simulation engine.



Progress: Milestone Completion Status

Milestone		Due Week	Completion Status	Remarks
#1	Literature Review & Scope Finalization	Week 2	100%	Completed: Comfort set δ defined (T: 20-24°C, RH: 40–60%, CO ₂ < 1000 ppm)
	Environment Setup and Initialization	Week 3	100%	Completed: Setup a communication between energyplus model & python script.
#2	Design a single-zone energyplus model and a controller that could read from sensors and write to actuators	Week 5	100%	Completed: Succesfully read data from temperature, humidity and AQ sensors and controlled actuators (VAV damper, Outdoor air damper)
#3	Design a 5-zone energyplus model	Week 6	100%	Completed: Succesfully designed an office space with 5 zones.
	Local controllers for each zone	Week 8	80%	Local individual controllers that inherit from a general controller. Yet to implement individul zone specific controls.
	Central AHU controller	Week 8	80%	Aggregates zone wise required mass flow rate and provide the supply air at constant temperature (12 or 14 degrees)
#4	Iterative development and tuning of the controller	Week 14	10%	Develop controls with variable supply air temperature, varying occupancy rates and estimators for the varying parameters
	Documentation, report of results	Week 14	0%	

Achieved Results - Simulation Platform Successfully Built and Running

A realistic 5-zone office served by a single VAV air-handling unit - built control-free so our Python controller drives every decision.

Annual Simulation - 52,560 timesteps (10-min resolution, full calendar year)

Location & climate:

- Colombo, Sri Lanka (6.9°N, 79.9°E) — hot & humid year-round
- Cooling design day: 33 °C dry-bulb / 27 °C wet-bulb → cooling-only problem, no heating needed
- Annual run, 10-minute timestep

The building:

- 5 thermal zones, each 5 × 8 × 3 m (40 m², 120 m³)
- Mixed-use loads: Open Office, Private Offices, Conference Room, Server Room (high heat: 60 W/m²), Reception
- South-facing glazing on every zone (SHGC 0.40)

The HVAC:

- Single VAV air loop: outdoor-air mixer → DX cooling coil (COP 3.5) → variable-speed fan
- 5 single-duct terminals, one per zone
- CO₂ contaminant balance enabled (IAQ tracking)

Why this model:

- No EMS, no built-in control logic - setpoints, airflows, and supply-air temperature are all left open for the Python controller to set every timestep.

HVAC sizing & control targets per zone

Zone	Cooling	Heating	Outdoor-air	Terminal
Zone 1	24 °C	18 °C	0.008	0.5
Zone 2	24 °C	18 °C	0.008	0.35
Zone 3	23 °C	18 °C	0.010	0.55
Zone 4	21 °C	18 °C	0.005	0.6
Zone 5	24 °C	18 °C	0.010	0.35

Internal loads per zone

Zone	Peo	Occupan	Lights	Equipme	Equipme	Infiltratio
Zone 1 —	8	Office	10	12	Office	0.0003
Zone 2 —	4	Office	9	10	Office	0.0003
Zone 3 —	12	Office	8	5	Office	0.0003
Zone 4 —	1	Office	6	60	Server	0.0003
Zone 5 —	5	Office	11	6	Office	0.0003

Achieved Results - Decentralised Control Architecture Demonstrated

Each zone runs its own independent P-controller that sees only its own measurements (T, w, RH, CO₂) - this is the core "decentralised" property.

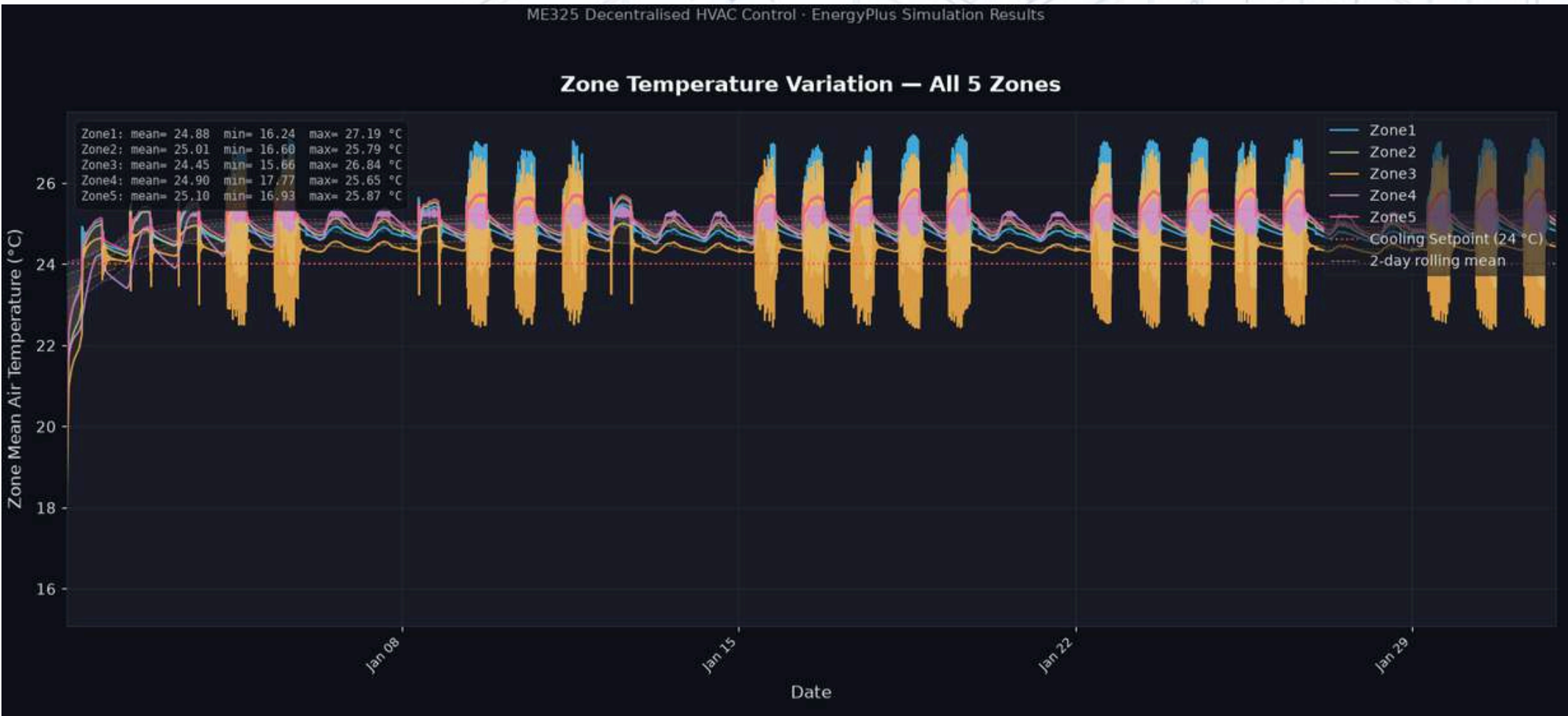
Output	Formula	Description
Supply mass flow (mdot)	$kp \times (T_{\text{zone}} - T_{\text{setpoint}}) \times \text{max_mdot}$	Proportional to temperature error, clamped to [0, max_mdot]
SAT request (t_sup_req)	12°C if RH > target, else 14°C	Humidity-aware binary request
Cooling setpoint	Fixed at 24°C (configurable)	Passed through to thermostat actuator
Outdoor Air Fraction	$\text{ota_frac} = (\text{CO2_max} - 400) / (800 - 400)$ $\text{oa_flow} = \text{oa_frac} \times \sum \text{mdot_zone}$	At least 0.15 of supply air is drawn from outside. {clamped to [0.15, 1]}

Zone 4 (Server Room) enforces a hard minimum airflow floor to prevent the server room from ever losing cooling - demonstrating the pluggable per-zone architecture.

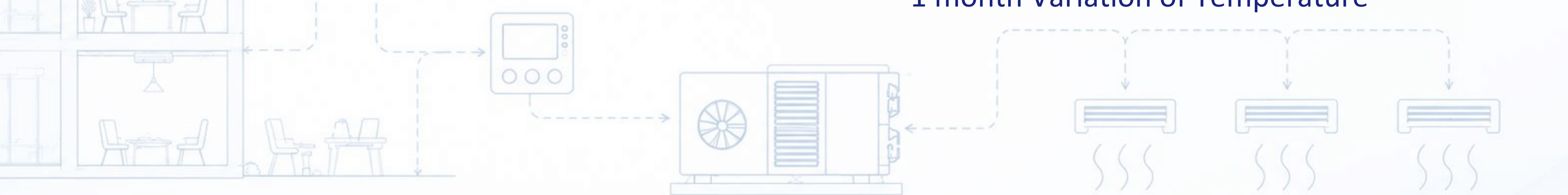
Quantitative Simulation Results (Full Year, 52,560 Timesteps)

Temperature

Zone	Use	Mean T (°C)	Std Dev	Max (°C)	% Time within ±1°C of setpoint
Zone 1	Open Office	24.87	0.61	27.24	82.0%
Zone 2	Private Offices	24.99	0.32	25.82	65.0%
Zone 3	Conference Room	24.44	0.72	26.88	80.7%
Zone 4	Server Room	24.89	0.29	25.69	63.1%
Zone 5	Reception	25.07	0.32	25.89	54.5%



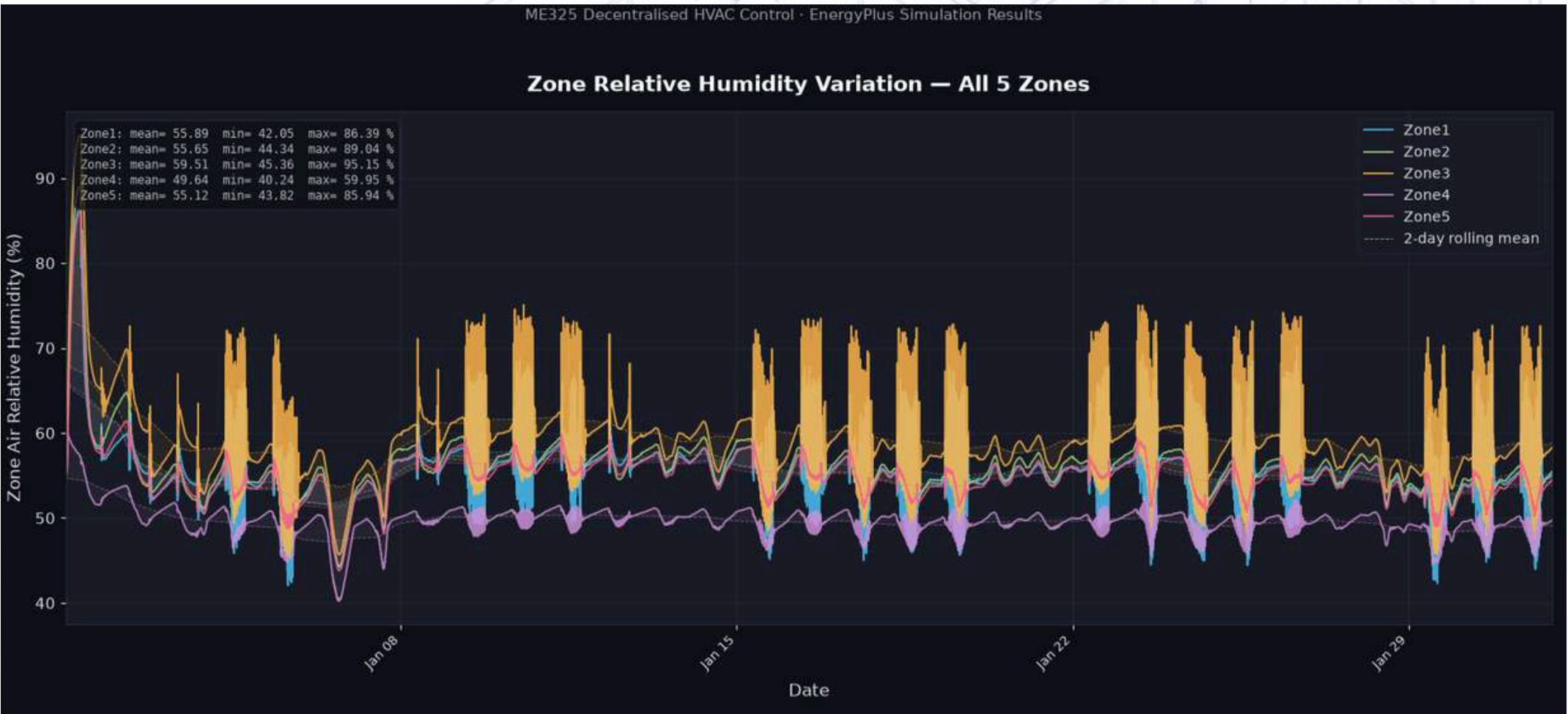
1 month Variation of Temperature



Quantitative Simulation Results (Full Year, 52,560 Timesteps)

Humidity

Zone	Mean RH (%)	Max RH (%)	% Time > 60%
Zone 1	57.1	86.4	14.6%
Zone 2	57.0	89.0	7.2%
Zone 3	60.9	95.2	56.7%
Zone 4	50.1	60.0	0.0%
Zone 5	56.4	85.9	3.3%

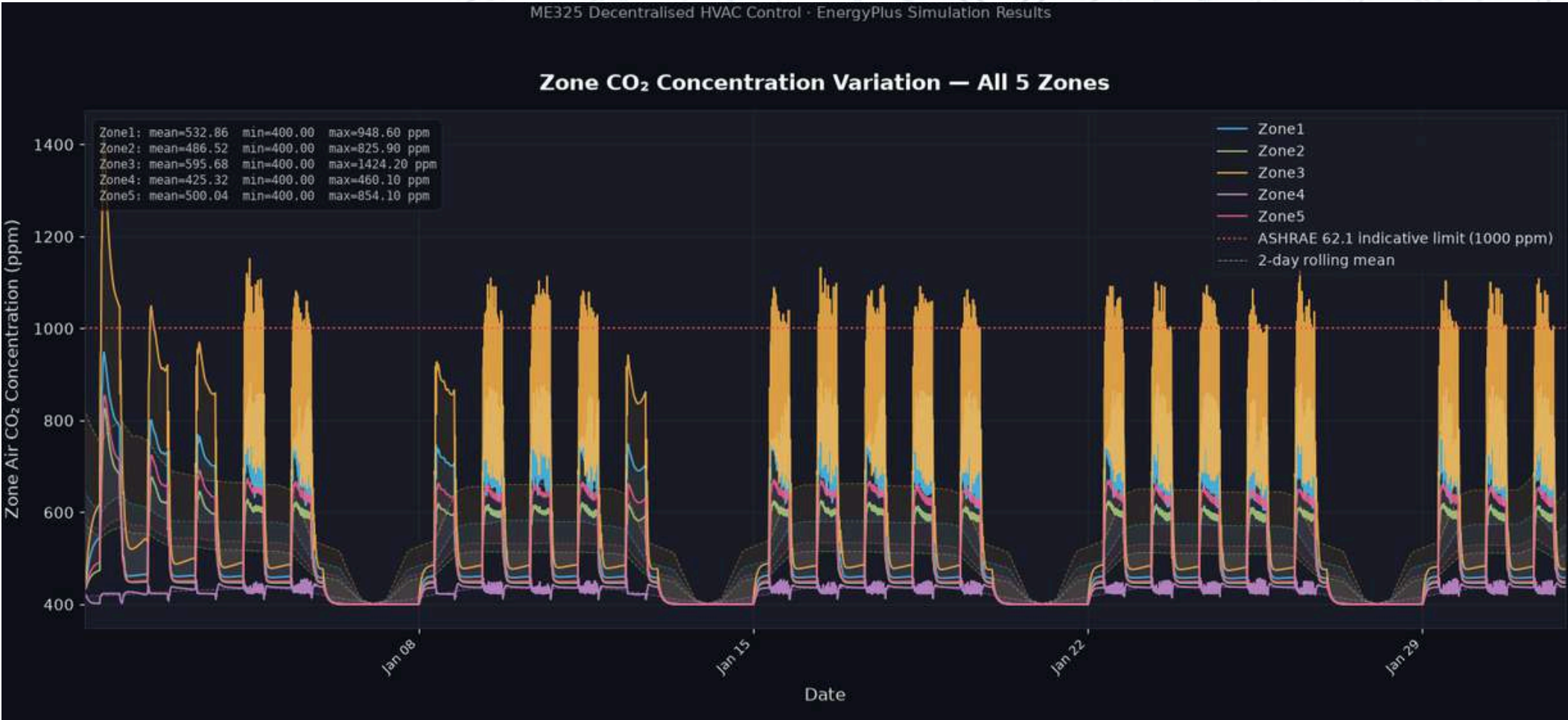


1 month Variation of Humidity

Quantitative Simulation Results (Full Year, 52,560 Timesteps)

Indoor Air Quality

Zone	People	Mean (ppm)	Max (ppm)	% > 1000 ppm	% > 800 ppm
Zone 1	8	531	949	0.0%	8.9%
Zone 2	4	484	826	0.0%	0.0%
Zone 3	12	592	1424	11.5%	19.6%
Zone 4	1	424	462	0.0%	0.0%
Zone 5	5	497	854	0.0%	0.0%



1 month Variation of Indoor Air Quality

Q&A

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